Technical Addendum to GREEN-MCS:

Analytical Pareto Knee-Point, G/G/1 Queuing Dynamics, and Algorithmic Complexity

1. Objective Space and Analytical Pareto Knee-Point Formulation

Unlike analytical methods (such as the Nash Bargaining Solution) which assume continuous and convex utility spaces, SDN hardware deployments represent a discrete, highly non-convex objective space. This is due to the discrete assignment of heterogeneous data-plane hardware profiles (low-power Zodiac FX vs. high-performance NEC PF5240) and the asymptotic behavior of G/G/1 queuing under heavy traffic loads.

To evaluate the multi-objective optimization without human bias, we first normalize the absolute objective values for energy (E_S) and delay (D_S) for any candidate set of failover switches S using Min-Max feature scaling:

$$\hat{E}_S = \frac{E_S - E_{min}}{E_{max} - E_{min}}, \quad \hat{D}_S = \frac{D_S - D_{min}}{D_{max} - D_{min}} \tag{1}$$

The hybrid cost function is then defined as a weighted linear combination using the routing policy parameter $\alpha \in [0, 1]$:

$$Score_S(\alpha) = \alpha \cdot \hat{E}_S + (1 - \alpha) \cdot \hat{D}_S \tag{2}$$

Instead of arbitrarily selecting an α value, the centralized framework dynamically calculates the **Pareto Knee-Point**—the exact geometrical threshold of maximum diminishing marginal returns [1]. As illustrated in **Fig. 1**, we define the anchor points of the Pareto frontier as the individual absolute minimums for energy and delay. In the normalized $[0, 1]$ plane, the X-axis represents $\hat{E}_S$ and the Y-axis represents $\hat{D}_S$.

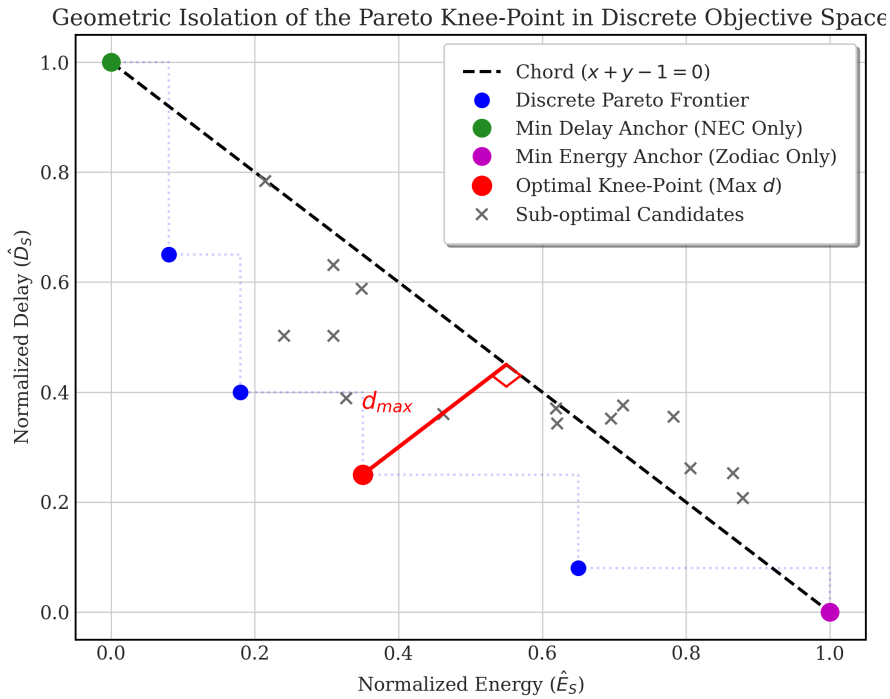

Figure 1: Geometric isolation of the optimal hardware configuration. The objective space is defined by the absolute minimums for delay (green anchor, pure NEC PF5240 deployment) and energy (magenta anchor, pure Zodiac FX deployment). Because the available hardware choices are binary, the valid configurations form a discrete set of sub-optimal candidates (grey crosses) and a discrete Pareto frontier (blue dots). The theoretical chord (dashed line) establishes the reference baseline $x + y - 1 = 0$. The centralized framework iteratively projects an orthogonal vector from each candidate onto this chord. The configuration corresponding to the Knee-Point (red dot) is autonomously selected by maximizing this orthogonal distance (d_{max}), identifying the strict mathematical threshold where further energy reductions would trigger an exponential SLA violation.

The theoretical worst-case trade-off is established by the chord connecting the ideal anchor points $(0, 1)$ and $(1, 0)$, which is geometrically defined by the straight line $\hat{E}_S + \hat{D}_S - 1 = 0$. The optimal Knee-Point is identified analytically by finding the physical hardware configuration that maximizes the orthogonal distance d_{max} to this chord:

$$d = \frac{|\hat{E}_S + \hat{D}_S - 1|}{\sqrt{2}} \tag{3}$$

By projecting this orthogonal distance, the centralized Traffic Engineering (TE) server executes a massive combinatorial sweep across discrete α values and traffic variance factors (σ). Using this exact mathematical formulation, it effortlessly discards sub-optimal candidates and autonomously locks onto the configuration that yields the highest energy savings without compromising the Service Level Agreement (SLA).

2. G/G/1 Queuing Dynamics and Stochastic Variance

To guarantee strict sub-50 ms SLAs—a Carrier-Grade standard established for Automatic Protection Switching [2]—under non-deterministic traffic, standard M/M/1 models (which assume Poisson arrivals and exponential service times) are insufficient. To accurately model the physical

delay under extreme micro-bursts and varying hardware forwarding architectures, we implement **Kingman's Approximation for G/G/1 queues** [4].

Kingman's formula elegantly separates the delay calculation into three distinct mathematical components: capacity utilization, traffic variance, and hardware speed. Before defining the total queuing delay (D_q), we establish its core independent variables.

First, the **Utilization Factor (ρ)** quantifies the fundamental load of the physical switch. It is the ratio of the aggregated incoming traffic post-failover (λ_{post_fail}) to the maximum processing capacity of the selected hardware (μ):

$$\rho = \frac{\lambda_{post_fail}}{\mu} \tag{4}$$

Second, the **Arrival Coefficient of Variation (c_a)** mathematically links the spatial traffic entropy directly to the queue stress. It is derived from the standard deviation of the traffic bursts (σ) relative to the mean arrival rate:

$$c_a = \frac{\sigma}{\lambda_{post_fail}} \tag{5}$$

By unifying these parameters with the hardware's intrinsic Service Coefficient of Variation (c_s), the expected queuing delay D_q at any physical switch is formulated as:

$$D_q = \underbrace{\left(\frac{\rho}{1-\rho}\right)}_{\text{Asymptotic Penalty}} \cdot \underbrace{\left(\frac{c_a^2 + c_s^2}{2}\right)}_{\text{Stochastic Multiplier}} \cdot \underbrace{\left(\frac{1}{\mu}\right)}_{\text{Base Service Time}} \tag{6}$$

Anatomical Breakdown of the Queuing Model:

- **The Asymptotic Penalty $\left(\frac{\rho}{1-\rho}\right)$:** This term dictates the non-linear explosion of delay (bufferbloat). As $\rho \to 1$, the queue length approaches infinity. Based on state-of-the-art SDN traffic engineering principles [3]), our centralized framework enforces a hard hardware-provisioning limit of $\rho \leq \mathbf{0.95}$. Operating beyond this limit invalidates SLA guarantees.
- **The Stochastic Multiplier $\left(\frac{c_a^2+c_s^2}{2}\right)$:** This term scales the delay based on the chaos of the environment.
  - c_a penalizes the network when incoming traffic (σ) is highly erratic (micro-bursts).
  - c_s differentiates the determinism of the data-plane hardware architectures. We consider a heterogeneous multi-tier hardware environment:
    - **High-Performance (NEC PF5240):** ASIC-based, highly deterministic packet processing ($\mu = 131$ Mpps, 118 W, $c_s = 0.2$).
    - **Low-Power (Zodiac FX):** Lightweight IoT-grade forwarding, highly irregular processing under stress ($\mu = 100$ kpps, 15 W, $c_s = 1.2$).
- **The Base Service Time $\left(\frac{1}{\mu}\right)$:** The theoretical minimum time required to forward a single packet in a perfect, empty network.

3. Worst-Case Provisioning and "SLA Immunity" Dynamics

To evaluate the network under maximum stochastic stress, our centralized framework integrates a **Worst-Case Failover** hardware provisioning logic. Before simulating the physical queuing delay (D_q) defined in Eq. 6, the algorithm must determine the maximum potential load any failover switch (Hero) will endure.

We define the absolute worst-case post-failover arrival rate ($\lambda^{max}_{post_fail}$) for a given switch by aggregating its steady-state baseline traffic (λ_{base}) with the heaviest single-link failure it is actively designated to rescue:

$$\lambda^{max}_{post_fail} = \lambda_{base} + \max_{\forall (u,v) \in \text{rescues}} \left(\lambda^{(u,v)}_{cut} \right) \tag{7}$$

By substituting $\lambda^{max}_{post_fail}$ into Eq. 4, the algorithm evaluates the worst-case Utilization Factor (ρ_{max}) for the default low-power hardware (Zodiac FX). If $\rho_{max} > 0.95$, the framework identifies an imminent collision with the *Asymptotic Penalty* wall. To prevent bufferbloat and guarantee the sub-50 ms SLA, the algorithm autonomously enforces a CAPEX upgrade. It physically provisions a high-performance ASIC switch (NEC PF5240) for that specific topological node to drastically increase service capacity (μ) and restore $\rho \leq 0.95$.

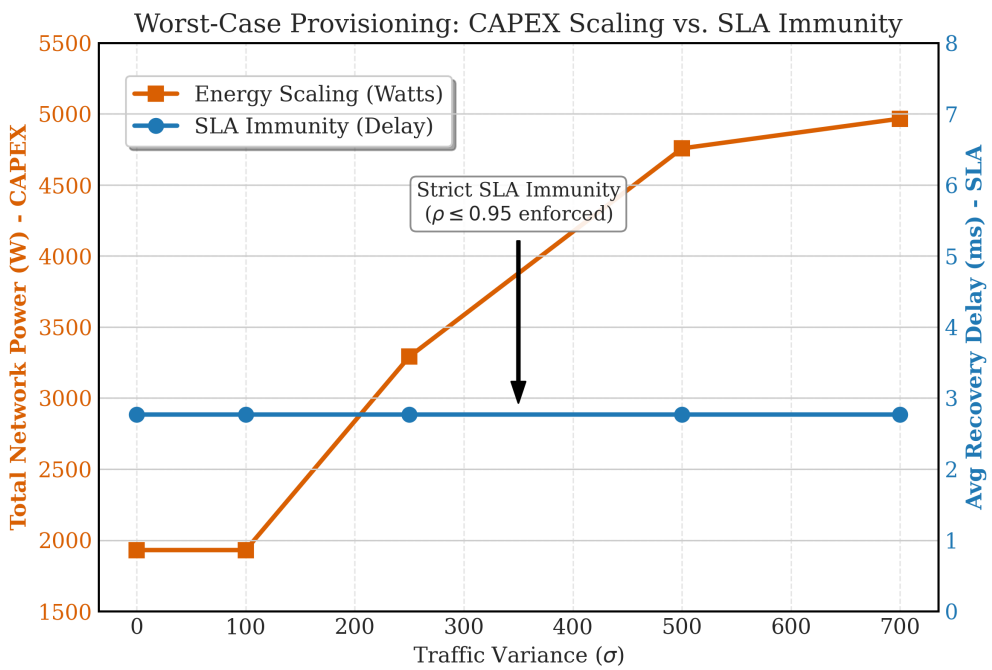

Figure 2: Dynamic hardware provisioning and SLA Immunity observed under increasing traffic variance (σ). By strictly bounding the Utilization Factor ($\rho \leq 0.95$), the framework prevents the Asymptotic Penalty from dominating the queuing delay. As traffic entropy increases, the system autonomously scales CAPEX (orange line) by deploying more high-performance NEC nodes. This strategic energy investment fully absorbs the stochastic variance, keeping the recovery delay (blue line) rigorously flat and well below the 50 ms carrier-grade threshold.

Empirical Results & Trade-off Collapse: As visually demonstrated in **Fig. 2**, because the framework mathematically shields the network from queue saturation under maximum stress, the *Stochastic Multiplier* is strictly contained. Consequently, the recovery delay becomes highly deterministic (locking at a flat ~ 2.77 ms for the evaluated topology across all σ variations).

This phenomenon fundamentally alters the multi-objective space. The traditional convex energy-delay trade-off collapses into a rigid "L-shaped" Pareto frontier. The orthogonal distance formula (Eq. 3) geometrically intersects this right angle consistently at $\alpha \approx 0.4$. This proves empirically that under worst-case SDN provisioning, the intelligent hybrid system absorbs 100% of the traffic entropy via Energy scaling (Watts) to guarantee absolute **SLA Immunity** (Milliseconds).

4. Exhaustive Complexity Analysis and K-Augmented Search

The proposed GREEN-MCS pipeline is designed to operate as a centralized application running on the SDN Controller (e.g., Ryu or a dedicated TE server). It computes the failover state and pushes the flow rules down to the data-plane switches. The algorithm executes in four sequential polynomial phases. Let V be the number of forwarding switches (nodes), E the number of fiber links (edges), F the total number of single-link failure scenarios (where $F = 2|E|$ for bidirectional links), N_α be the number of discrete routing policy weights evaluated (e.g., step intervals between 0 and 1), and N_σ be the number of stochastic traffic variance scenarios simulated.

1. **Failure Impact Analysis** – $\mathcal{O}(F \cdot (V \log V + E))$**:** The centralized controller iterates through all F failure scenarios. For each failure, it executes Dijkstra's algorithm to identify affected destinations. This phase establishes the reachability baseline under stress.

2. **Hero Switch Validation** – $\mathcal{O}(F \cdot V \cdot (V \log V + E))$**:** For every link failure, the system evaluates all V potential "Hero" switches (failover anchors). Validation requires checking path existence from the source of the cut to the candidate switch, and from the candidate to all affected destinations.

3. **K-Augmented Greedy Search** – $\mathcal{O}(F \cdot V + V^2)$**:** The foundational failover problem is modeled as an NP-Hard Set Cover. A standard greedy heuristic yields a minimal cardinality set of switches (S_{base}) in $\mathcal{O}(F \cdot V)$. However, the minimal cardinality solution is rarely Pareto-optimal. To escape local optima, we perform a **Quadratic Neighborhood Expansion**. By generating all combinations of $S_{base} + k$ (for $k \in \{1, 2\}$), we expand the solution space to $1 + |V| + \binom{|V|}{2}$ candidate sets. This tests the hypothesis that routing traffic through a marginal amount of additional passive hardware reduces the global propagation and queuing delay super-linearly.

4. **Pareto Tribunal & Multi-Objective Sweep** – $\mathcal{O}(N_\alpha \cdot N_\sigma \cdot V^2 \cdot F)$**:** The final optimization sweep evaluates the expanded candidate sets across N_α priority weights and N_σ traffic variances. Each evaluation involves calculating the Kingman G/G/1 queuing delay and hardware power states for all F failures.

Total Complexity Summary: The overall complexity is dominated by the evaluation of augmented sets and path validations, resulting in an asymptotic upper bound of $\mathcal{O}(V^2 \cdot E \cdot V \log V)$. While seemingly heavy, it is computationally trivial for a modern centralized server. Given that real-world carrier-grade topologies (e.g., Abilene, Nobel-Germany, Germany50) possess a relatively small number of logical switches ($V < 100$), this approach scales efficiently. This validates its readiness to be deployed as a central Traffic Engineering module in Software-Defined Wide Area Networks (SD-WAN), efficiently orchestrating simple data-plane switches.

References

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